

CS 550 – PA4
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Output File

Indexing Server and 5 Peers are started with Peers 1 and 4 having same files.

- Starting the Indexing Server: `./run_server.sh`

```
(base) fatimamariyam@Fatimas-MacBook-Air p2p-pa4 % ./run_server.sh
#####
Running the Server script...
#####
***** INDEXING SERVER STARTED *****
```

- Starting peer 1 `./run_peer.sh`

```
(base) fatimamariyam@Fatimas-MacBook-Air p2p-pa4 % ./run_peer.sh
#####
Running the Peer script...
#####
Enter the port number for this peer
1001
Enter the directory path to register with the Indexing Server
/Users/fatimamariyam/Documents/HW/AOS/p2p-pa4/SHARED_DIR/Folder1
***** PEER CLIENT STARTED *****
1. Work
2. Work Parallel
2
Give a Peer ID to this Peer
1
Registering file file1_5 in Indexing Server of size = 141312
Registering file file1_2 in Indexing Server of size = 70656
Registering file file1_11 in Indexing Server of size = 92160
Registering file file1_16 in Indexing Server of size = 204800
Registering file file1_3 in Indexing Server of size = 164864
Registering file file1_4 in Indexing Server of size = 230400
Registering file file1_10 in Indexing Server of size = 219136
Registering file file1_15 in Indexing Server of size = 211968
Registering file file1_12 in Indexing Server of size = 78848
Registering file file1_1 in Indexing Server of size = 158720
Registering file file1_6 in Indexing Server of size = 245760
Registering file file1_8 in Indexing Server of size = 179200
Registering file file1_13 in Indexing Server of size = 198656
Registering file file1_14 in Indexing Server of size = 226304
Registering file file1_9 in Indexing Server of size = 126976
Registering file file1_7 in Indexing Server of size = 198656
Please make a choice
1. Query File
2. Un-register File
3. Exit
```

- Starting peer 2 `./run_peer.sh`

```
#####
Running the Peer script...
```

```
#####
Enter the port number for this peer
1002
Enter the directory path to register with the Indexing Server
/Users/fatimamariyam/Documents/HW/AOS/p2p-pa4/SHARED_DIR/Folder1
***** PEER CLIENT STARTED *****
1. Work
2. Work Parallel
2
Give a Peer ID to this Peer
2
Registering file file1_5 in Indexing Server of size = 141312
Registering file file1_2 in Indexing Server of size = 70656
Registering file file1_11 in Indexing Server of size = 92160
Registering file file1_16 in Indexing Server of size = 204800
Registering file file1_3 in Indexing Server of size = 164864
Registering file file1_4 in Indexing Server of size = 230400
Registering file file1_10 in Indexing Server of size = 219136
Registering file file1_15 in Indexing Server of size = 211968
Registering file file1_12 in Indexing Server of size = 78848
Registering file file1_1 in Indexing Server of size = 158720
Registering file file1_6 in Indexing Server of size = 245760
Registering file file1_8 in Indexing Server of size = 179200
Registering file file1_13 in Indexing Server of size = 198656
Registering file file1_14 in Indexing Server of size = 226304
Registering file file1_9 in Indexing Server of size = 126976
Registering file file1_7 in Indexing Server of size = 198656
Please make a choice
1. Query File
2. Un-register File
3. Exit
```

- Starting peer 3 `./run_peer.sh`

```
#####
Running the Peer script...
#####
Enter the port number for this peer
1003
Enter the directory path to register with the Indexing Server
/Users/fatimamariyam/Documents/HW/AOS/p2p-pa4/SHARED_DIR/Folder3
***** PEER CLIENT STARTED *****
1. Work
2. Work Parallel
2
Give a Peer ID to this Peer
3
Registering file file3_1 in Indexing Server of size = 238592
Registering file file3_2 in Indexing Server of size = 135168
Registering file file3_5 in Indexing Server of size = 146432
Registering file file3_4 in Indexing Server of size = 141312
Registering file file3_3 in Indexing Server of size = 252928
Please make a choice
1. Query File
2. Un-register File
3. Exit
```

- Starting peer 4 `./run_peer.sh`

```
#####
Running the Peer script...
#####
Enter the port number for this peer
1004
Enter the directory path to register with the Indexing Server
/Users/fatimamariyam/Documents/HW/AOS/p2p-pa4/SHARED_DIR/folder1
***** PEER CLIENT STARTED *****
1. Work
2. Work Parallel
2
Give a Peer ID to this Peer
4
```

```

Registering file file1_5 in Indexing Server of size = 262144
Registering file file1_2 in Indexing Server of size = 89088
Registering file file1_11 in Indexing Server of size = 212992
Registering file file1_16 in Indexing Server of size = 114688
Registering file file1_18 in Indexing Server of size = 180224
Registering file file1_3 in Indexing Server of size = 200704
Registering file file1_4 in Indexing Server of size = 238592
Registering file file1_19 in Indexing Server of size = 83968
Registering file file1_17 in Indexing Server of size = 124928
Registering file file1_10 in Indexing Server of size = 177152
Registering file file1_15 in Indexing Server of size = 248832
Registering file file1_12 in Indexing Server of size = 186368
Registering file file1_1 in Indexing Server of size = 133120
Registering file file1_6 in Indexing Server of size = 249856
Registering file file1_8 in Indexing Server of size = 107520
Registering file file1_13 in Indexing Server of size = 257024
Registering file file1_14 in Indexing Server of size = 136192
Registering file file1_9 in Indexing Server of size = 178176
Registering file file1_7 in Indexing Server of size = 228352
Please make a choice

```

1. Query File
2. Un-register File
3. Exit

- **Starting peer 5** ./run_peer.sh

```

#####
Running the Peer script...
#####
Enter the port number for this peer
1005
Enter the directory path to register with the Indexing Server
/Users/fatimamariyam/Documents/HW/AOS/p2p-pa4/SHARED_DIR/folder4
***** PEER CLIENT STARTED *****
1. Work
2. Work Parallel
2
Give a Peer ID to this Peer
5
Registering file file4_9 in Indexing Server of size = 71680
Registering file file4_7 in Indexing Server of size = 151552
Registering file file4_6 in Indexing Server of size = 146432
Registering file file4_1 in Indexing Server of size = 120832
Registering file file4_8 in Indexing Server of size = 256000
Registering file file4_11 in Indexing Server of size = 128000
Registering file file4_10 in Indexing Server of size = 143360
Registering file file4_4 in Indexing Server of size = 173056
Registering file file4_3 in Indexing Server of size = 165888
Registering file file4_2 in Indexing Server of size = 250880
Registering file file4_5 in Indexing Server of size = 92160
Registering file file4_12 in Indexing Server of size = 242688
Registering file file4_14 in Indexing Server of size = 108544
Registering file file4_13 in Indexing Server of size = 80896
Please make a choice
1. Query File
2. Un-register File
3. Exit

```

```

X java                               X java
(base) fatimamariyam@Fatimas-MacBook-Air p2p-pa4 % ./run_server.sh
#####
Running the Server script...
#####
***** INDEXING SERVER STARTED *****
#####
THE UPDATED INDEX at 2024/04/04 17:07:21
file1_5 => [[file1_5, 1, 1001, /Users/fatimamariyam/Documents/HW/AOS/p2p-pa4/SHARED_DIR/folder1, 262144
]]

X java                               X java
(base) fatimamariyam@Fatimas-MacBook-Air p2p-pa4 % ./run_peer.sh
#####
Running the Peer script...
#####
Enter the port number for this peer
1002
Enter the directory path to register with the Indexing Server
/Users/fatimamariyam/Documents/HW/AOS/p2p-pa4/SHARED_DIR/folder2
***** PEER CLIENT STARTED *****
1. Work
2. Work Parallel
2
Give a Peer ID to this Peer
2
Registering file file2_5 in Indexing Serverof size = 102400
Registering file file2_2 in Indexing Serverof size = 116736
Registering file file2_3 in Indexing Serverof size = 147456
Registering file file2_4 in Indexing Serverof size = 126976
Registering file file2_1 in Indexing Serverof size = 159744
Registering file file2_6 in Indexing Serverof size = 151552
Registering file file2_8 in Indexing Serverof size = 121856
Registering file file2_9 in Indexing Serverof size = 78848

X java                               X java
(base) fatimamariyam@Fatimas-MacBook-Air p2p-pa4 % ./run_peer.sh
#####
Running the Peer script...
#####
Enter the port number for this peer
1003
Enter the directory path to register with the Indexing Server
/Users/fatimamariyam/Documents/HW/AOS/p2p-pa4/SHARED_DIR/folder3
***** PEER CLIENT STARTED *****
1. Work
2. Work Parallel
2
Give a Peer ID to this Peer
3
Registering file file3_13 in Indexing Serverof size = 257024
Registering file file3_14 in Indexing Serverof size = 99328
Registering file file3_15 in Indexing Serverof size = 176128
Registering file file3_12 in Indexing Serverof size = 83968
Registering file file3_8 in Indexing Serverof size = 244736
Registering file file3_6 in Indexing Serverof size = 108472
Registering file file3_5 in Indexing Serverof size = 113664
Registering file file3_7 in Indexing Serverof size = 143360
Registering file file3_9 in Indexing Serverof size = 137216

X java                               X java
(base) fatimamariyam@Fatimas-MacBook-Air p2p-pa4 % ./run_peer.sh
#####
Running the Peer script...
#####
Enter the port number for this peer
1004
Enter the directory path to register with the Indexing Server
/Users/fatimamariyam/Documents/HW/AOS/p2p-pa4/SHARED_DIR/folder1
***** PEER CLIENT STARTED *****
1. Work
2. Work Parallel
2
Give a Peer ID to this Peer
4
Registering file file1_5 in Indexing Serverof size = 262144
Registering file file1_2 in Indexing Serverof size = 89088
Registering file file1_11 in Indexing Serverof size = 212992
Registering file file1_16 in Indexing Serverof size = 114688
Registering file file1_18 in Indexing Serverof size = 180224
Registering file file1_3 in Indexing Serverof size = 200704
Registering file file1_4 in Indexing Serverof size = 238592

X java                               X java
Last login: Thu Apr  4 15:43:16 on ttys006
(base) fatimamariyam@Fatimas-MacBook-Air p2p-pa4 % ./run_peer.sh
#####
Running the Peer script...
#####
Enter the port number for this peer
1005
Enter the directory path to register with the Indexing Server
/Users/fatimamariyam/Documents/HW/AOS/p2p-pa4/SHARED_DIR/folder4
***** PEER CLIENT STARTED *****
1. Work
2. Work Parallel
2
Give a Peer ID to this Peer
5
Registering file file4_9 in Indexing Serverof size = 71680
Registering file file4_7 in Indexing Serverof size = 151552
Registering file file4_6 in Indexing Serverof size = 146432
Registering file file4_1 in Indexing Serverof size = 120832

```

- Requesting “file1_11” from Peer 2 - present in Peer 1 and 4

Enter name of the file you want to look for at indexing server:

file1_11

List of peer IDs holding the requested file:

Peer-ID: 1

Peer-ID: 4

Number of chunks = 4

File download in process...

File download in process...

Chunk 1 downloaded successfully

Chunk 0 downloaded successfully

File download in process...

File download in process...

Chunk 2 downloaded successfully

Chunk 3 downloaded successfully

Chunk 0 downloaded successfully

Chunk 1 downloaded successfully

Chunk 2 downloaded successfully

Chunk 3 downloaded successfully

File downloaded successfully from multiple peers.

Integrity check passed: Downloaded file matches the original file.

1. Query File

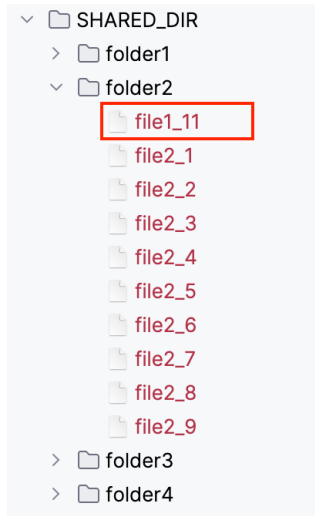
2. Un-register File

3. Exit

```

Registering file file2_8 in Indexing Server of size = 121856
Registering file file2_9 in Indexing Server of size = 78848
Registering file file2_7 in Indexing Server of size = 225280
Please make a choice
1. Query File
2. Un-register File
3. Exit
1
Enter name of the file you want to look for at indexing server:
file1_11
List of peer IDs holding the requested file:
Peer-ID: 1
Peer-ID: 4
Number of chunks = 4
File download in process...
File download in process...
Chunk 1 downloaded successfully
Chunk 0 downloaded successfully
File download in process...
File download in process...
Chunk 2 downloaded successfully
Chunk 3 downloaded successfully
Chunk 0 downloaded successfully
Chunk 1 downloaded successfully
Chunk 2 downloaded successfully
Chunk 3 downloaded successfully
File downloaded successfully from multiple peers.
Integrity check passed: Downloaded file matches the original file.
1. Query File
2. Un-register File
3. Exit

```



- Size of downloaded file in folder2 = size of original file in folder1

```

(base) fatimamariyam@Fatimas-MacBook-Air p2p-pa4 % ls -lh SHARED_DIR/folder1
total 6712
-rw-r--r--@ 1 fatimamariyam staff 130K Apr  4 16:48 file1_1
-rw-r--r--@ 1 fatimamariyam staff 173K Apr  4 16:48 file1_10
-rw-r--r--@ 1 fatimamariyam staff 208K Apr  4 16:48 file1_11
-rw-r--r--@ 1 fatimamariyam staff 182K Apr  4 16:48 file1_12
-rw-r--r--@ 1 fatimamariyam staff 251K Apr  4 16:48 file1_13
-rw-r--r--@ 1 fatimamariyam staff 133K Apr  4 16:48 file1_14
-rw-r--r--@ 1 fatimamariyam staff 243K Apr  4 16:48 file1_15
-rw-r--r--@ 1 fatimamariyam staff 112K Apr  4 16:48 file1_16
-rw-r--r--@ 1 fatimamariyam staff 122K Apr  4 16:48 file1_17
-rw-r--r--@ 1 fatimamariyam staff 176K Apr  4 16:48 file1_18
-rw-r--r--@ 1 fatimamariyam staff  82K Apr  4 16:48 file1_19
-rw-r--r--@ 1 fatimamariyam staff  87K Apr  4 16:48 file1_2
-rw-r--r--@ 1 fatimamariyam staff 196K Apr  4 16:48 file1_3
-rw-r--r--@ 1 fatimamariyam staff 233K Apr  4 16:48 file1_4
-rw-r--r--@ 1 fatimamariyam staff 256K Apr  4 16:48 file1_5
-rw-r--r--@ 1 fatimamariyam staff 244K Apr  4 16:48 file1_6
-rw-r--r--@ 1 fatimamariyam staff 223K Apr  4 16:48 file1_7
-rw-r--r--@ 1 fatimamariyam staff 105K Apr  4 16:48 file1_8
-rw-r--r--@ 1 fatimamariyam staff 174K Apr  4 16:48 file1_9

```

```
(base) fatimamariyam@Fatimas-MacBook-Air p2p-pa4 % ls -lh SHARED_DIR/folder2
total 2832
-rw-r--r--  1 fatimamariyam  staff   208K Apr  4 17:23 file1_11
-rw-r--r--@ 1 fatimamariyam  staff  156K Apr  4 16:48 file2_1
-rw-r--r--@ 1 fatimamariyam  staff  114K Apr  4 16:48 file2_2
-rw-r--r--@ 1 fatimamariyam  staff  144K Apr  4 16:48 file2_3
-rw-r--r--@ 1 fatimamariyam  staff  124K Apr  4 16:48 file2_4
-rw-r--r--@ 1 fatimamariyam  staff  100K Apr  4 16:48 file2_5
-rw-r--r--@ 1 fatimamariyam  staff  148K Apr  4 16:48 file2_6
-rw-r--r--@ 1 fatimamariyam  staff  220K Apr  4 16:48 file2_7
-rw-r--r--@ 1 fatimamariyam  staff  119K Apr  4 16:48 file2_8
-rw-r--r--@ 1 fatimamariyam  staff   77K Apr  4 16:48 file2_9
```

- Deleting file from Peer2

```
1. Query File
2. Un-register File
3. Exit
2
Enter the file name which you want to delete
file1_11
File deleted Successfully.
Do you want to delete more files? (Yes/No)
no
1. Query File
2. Un-register File
3. Exit
```

- Indexing server logging

```
52], [file1_10, 4, 1004, /Users/fatimamariyam/Documents/HW/AOS/p2p-pa4/SHARED_DIR/folder1, 177152]]
file1_11 => [[file1_11, 1, 1001, /Users/fatimamariyam/Documents/HW/AOS/p2p-pa4/SHARED_DIR/folder1, 2129
92], [file1_11, 4, 1004, /Users/fatimamariyam/Documents/HW/AOS/p2p-pa4/SHARED_DIR/folder1, 212992]]
file1_12 => [[file1_12, 1, 1001, /Users/fatimamariyam/Documents/HW/AOS/p2p-pa4/SHARED_DIR/folder1, 1863
68], [file1_12, 4, 1004, /Users/fatimamariyam/Documents/HW/AOS/p2p-pa4/SHARED_DIR/folder1, 186368]]
file1_13 => [[file1_13, 1, 1001, /Users/fatimamariyam/Documents/HW/AOS/p2p-pa4/SHARED_DIR/folder1, 2570
24], [file1_13, 4, 1004, /Users/fatimamariyam/Documents/HW/AOS/p2p-pa4/SHARED_DIR/folder1, 257024]]
file1_14 => [[file1_14, 1, 1001, /Users/fatimamariyam/Documents/HW/AOS/p2p-pa4/SHARED_DIR/folder1, 1361
92], [file1_14, 4, 1004, /Users/fatimamariyam/Documents/HW/AOS/p2p-pa4/SHARED_DIR/folder1, 136192]]
file1_15 => [[file1_15, 1, 1001, /Users/fatimamariyam/Documents/HW/AOS/p2p-pa4/SHARED_DIR/folder1, 2488
32], [file1_15, 4, 1004, /Users/fatimamariyam/Documents/HW/AOS/p2p-pa4/SHARED_DIR/folder1, 248832]]
file1_16 => [[file1_16, 1, 1001, /Users/fatimamariyam/Documents/HW/AOS/p2p-pa4/SHARED_DIR/folder1, 1146
88], [file1_16, 4, 1004, /Users/fatimamariyam/Documents/HW/AOS/p2p-pa4/SHARED_DIR/folder1, 114688]]
file4_8 => [[file4_8, 5, 1005, /Users/fatimamariyam/Documents/HW/AOS/p2p-pa4/SHARED_DIR/folder4, 256000
]]
file2_9 => [[file2_9, 2, 1002, /Users/fatimamariyam/Documents/HW/AOS/p2p-pa4/SHARED_DIR/folder2, 78848]
]
file4_7 => [[file4_7, 5, 1005, /Users/fatimamariyam/Documents/HW/AOS/p2p-pa4/SHARED_DIR/folder4, 151552
]]
file4_9 => [[file4_9, 5, 1005, /Users/fatimamariyam/Documents/HW/AOS/p2p-pa4/SHARED_DIR/folder4, 71680]
]
#####
]
```